

Fake News Classification and Explainability

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Abstract

Fake news detection using transformer-based classifiers often achieves near-perfect performance on benchmark datasets; however, such performance may reflect reliance on dataset-specific artifacts rather than genuine language understanding. This project investigates this issue through two complementary components: model auditing using explainability methods and the generation of human-readable explanations. First, SHAP and LIME were applied to DistilBERT and RoBERTa classifiers trained on the ISOT Fake News dataset. Although the models achieved accuracies of 99.97% and 99.88%, respectively, the explanations revealed two sources of data leakage: a wire-service formatting artifact and a strong correlation between subject categories and class labels. After controlling for these artifacts, DistilBERT accuracy decreased to 21.91%, demonstrating the substantial impact of dataset-specific signals on the original performance. Second, a pipeline was developed using SHAP attributions and Google Gemini to generate natural-language explanations for predictions on the LIAR dataset. The explanations were evaluated using token coverage, directional accuracy, and semantic consistency, achieving 78% token coverage, 100% directional accuracy on checkable cases, and a cosine similarity of 0.88 across repeated generations. A small user study with 12 participants found that the explanations received average ratings of 3.90/5 for clarity and 3.62/5 for increasing trust in predictions, while 83.3% of participants preferred plain-language explanations over raw attribution scores. Overall, the results demonstrate that explainability methods can be used both to identify hidden weaknesses in high-performing classification models and to communicate model decisions in a more accessible form.

1. Introduction

The increasing prevalence of misinformation and AI-generated content has made it more challenging to distinguish credible news from fabricated information. This project develops a fake news detection system using fine-tuned transformer-based models and focuses on improving the interpretability of their predictions. Although explainability methods such as SHAP and LIME can identify the features that contribute to a model's decision, their raw attribution scores can be difficult for non-technical users to interpret. To address this limitation, the proposed system combines transformer-based classification with SHAP/LIME feature attributions and a large language model (LLM) to generate concise, natural-language explanations of model predictions.

During the evaluation of the classification models, explainability analysis also revealed an important limitation in the ISOT Fake News dataset. The initial models achieved unexpectedly high performance, prompting an investigation into the features driving their predictions. This analysis identified two independent sources of data leakage that contributed substantially to the observed performance. Based on these findings, the project consists of two main components.

Part 1 – Dataset Leakage Analysis on ISOT – SHAP and LIME were used to examine model predictions, with particular attention to disagreements and false-positive cases. The analysis identified a wire-service dateline as a major dataset artifact influencing the predictions. After controlling for this artifact, a second and larger source of leakage was identified through the strong correlation between subject categories and class labels. These results demonstrate that explainability techniques can be useful not only for interpreting predictions but also for identifying potential problems in datasets and auditing model behavior.

Part 2 – LLM-Generated Explanations on LIAR – The second component uses the LIAR dataset, which does not exhibit the same leakage patterns identified in ISOT. A pipeline was developed in which SHAP feature attributions are provided to Google Gemini to generate concise natural-language explanations of model predictions. Since an LLM-generated explanation may appear reasonable without accurately reflecting the underlying model, the explanations are evaluated using three measures of faithfulness: token coverage, directional accuracy, and semantic consistency. These measures assess whether the generated explanations remain grounded in the features that influenced the model's prediction.

Together, these components examine explainability from two perspectives: model auditing and user-facing interpretation. The project demonstrates how explainability techniques can help identify hidden sources of model performance while also providing a practical approach for communicating model decisions to users in a more understandable form.

2. Related Work

Fake news detection has evolved from traditional machine learning approaches, such as TF-IDF combined with Support Vector Machines (SVM) and Logistic Regression, to deep learning and transformer-based models, including BERT, DistilBERT, and RoBERTa (Khan et al., 2021; Kitanovski et al., 2023). Transformer-based approaches have generally achieved strong performance on benchmark datasets, though rankings vary across studies and datasets. Raza et al. (2024) compared fine-tuned BERT-family encoder models against autoregressive large language models on fake news classification and found that the fine-tuned encoder models generally outperformed LLMs on the classification task itself, while LLMs showed greater robustness to text perturbations — motivating this project's choice to use fine-tuned encoder classifiers (DistilBERT, RoBERTa) rather than prompting an LLM directly for classification.

LIME (Ribeiro et al., 2016) and SHAP (Lundberg & Lee, 2017) are widely used methods for explaining individual predictions in text classification. These methods identify the features that contribute to a model's decision, but their raw feature-importance scores and visualizations can be difficult for non-expert users to interpret (Gongane et al., 2024). Patel et al. (2025) applied LIME and SHAP directly to fine-tuned DistilBERT and RoBERTa misinformation classifiers, producing token-level and global attribution outputs to support transparency. This project extends that approach by translating such attributions into natural-language explanations using an LLM, and evaluating whether the resulting explanations accurately reflect the underlying model's reasoning or simply provide plausible-sounding interpretations — a question closely related to faithfulness evaluation methods developed in areas such as abstractive summarization (Maynez et al., 2020; Goodrich et al., 2019).

Another relevant area is shortcut learning, where machine learning models rely on superficial or unintended patterns instead of features that are genuinely relevant to the target task (Geirhos et al., 2020). This phenomenon is particularly important when evaluating models on benchmark datasets, since high test accuracy may result from dataset-specific artifacts rather than robust generalization. In this project, explainability methods are used to investigate this issue in the ISOT dataset, while the LLM-based explanation pipeline is evaluated separately on the LIAR dataset.

3. Datasets and Models

3.1 Datasets

ISOT Fake and Real News Dataset – The ISOT dataset contains 44,898 full-length news articles labeled as real or fake. Each article includes the title, body text, subject category, and publication date. The dataset was initially selected because its full-length articles provide sufficient text for attribution-based analysis

and explanation. However, as discussed in Section 4, the dataset was found to contain two independent sources of substantial label leakage.

LIAR Dataset – The LIAR dataset contains 12,791 short political statements collected from PolitiFact. The original dataset uses six truthfulness categories: *pants-fire*, *false*, *barely-true*, *half-true*, *mostly-true*, and *true*. For this project, the labels were converted to a binary classification task by defining *pants-fire* and *false* as **fake**, and *mostly-true* and *true* as **real**. The ambiguous middle categories, *barely-true* and *half-true*, were excluded to obtain clearer class boundaries. LIAR was used as the primary dataset for evaluating the explainability and LLM-based explanation pipeline because it does not exhibit the same dateline or subject-category leakage identified in ISOT.

3.2 Models

Three classification approaches were trained and evaluated on both datasets: a TF-IDF with Logistic Regression baseline, fine-tuned DistilBERT, and fine-tuned RoBERTa. The TF-IDF baseline used a maximum of 20,000 features, including unigram and bigram features, with English stopword removal.

For the transformer models, DistilBERT (**distilbert-base-uncased**) and RoBERTa (**roberta-base**) were fine-tuned using a learning rate of $2e-5$, the AdamW optimizer, and a batch size of 16. Training was performed for a maximum of five epochs with early stopping using validation F1-score and a patience of two epochs. The maximum sequence length was set to 256 tokens for ISOT and 64 tokens for LIAR. A sequence length of 512 was initially tested for ISOT but did not provide an improvement in accuracy while requiring approximately 2–3 times more computational resources. Therefore, 256 tokens was selected as a more efficient configuration. The shorter sequence length for LIAR reflects the dataset's short, statement-level format.

3.3 Baseline Model Comparison

Table 1 reports test-set performance on ISOT under a standard random 70/15/15 stratified split. Table 2 reports the same for LIAR.

Model	Accuracy	F1	Train Time
TF-IDF + Logistic Regression	98.60%	0.984	~25s
DistilBERT	99.98%	0.9998	~28 min
RoBERTa	99.88%	0.9987	~68 min

Table 1. Model comparison on ISOT (random split).

Model	Accuracy
TF-IDF + Logistic Regression	62.41%
DistilBERT	68.86%
RoBERTa	60.76%

Table 2. Model comparison on LIAR (predefined split, strict binary collapsing).

The near-perfect accuracy achieved by the transformer models on the ISOT dataset (>99.8%), compared with the substantially lower and more realistic accuracy observed on LIAR (60–69%), provided the initial indication that further investigation of the ISOT results was necessary. This observation motivated the dataset leakage analysis presented in Section 4.

4. Part 1 – Dataset Leakage Analysis on ISOT

4.1 Motivation

The initial DistilBERT model achieved 99.98% test accuracy on the ISOT dataset. Such near-perfect performance is unusually high for a task requiring meaningful understanding of misinformation and therefore motivated further investigation. Rather than treating the result as evidence of strong generalization, SHAP and LIME were used as diagnostic tools to examine which features were influencing the model's predictions.

4.2 Discovering the Reuters Dateline Artifact

To investigate the source of the high performance, cases where the TF-IDF baseline and DistilBERT disagreed were examined using SHAP and LIME, with particular attention to false positives. The analysis identified the word “Reuters” as a frequently occurring high-weight feature in both explanation methods. Its presence consistently contributed toward the real class.

A dataset-level analysis confirmed that this pattern was a significant artifact. Approximately 98.7% of real articles in the ISOT test set begin with a wire-service dateline following a pattern such as “CITY (Reuters) -”, whereas none of the fake articles contain this pattern. Furthermore, false-positive predictions from both DistilBERT and RoBERTa occurred exclusively among the small proportion of real articles that lacked the dateline. This provided strong evidence that the dateline was being used as a shortcut for distinguishing between the two classes.

4.3 Controlled Experiment: Removing the Shortcut

To measure the models' reliance on the dateline, the identified pattern was removed from the test-set articles and the already-trained models were evaluated again without retraining. The modified text was also manually inspected to ensure that the removal did not introduce noticeable formatting artifacts.

Model	Original Accuracy	Stripped Accuracy	Drop
TF-IDF + Logistic Regression	98.60%	98.05%	-0.55 pts
DistilBERT	99.98%	72.83%	-27.16 pts
RoBERTa	99.88%	45.77%	-54.11 pts

Table 3. Accuracy on dateline-stripped test text (no retraining).

The TF-IDF model was only minimally affected, suggesting that its predictions were based on a more distributed set of textual features. In contrast, DistilBERT experienced a substantial performance decrease, indicating significant reliance on the dateline artifact while retaining some predictive signal. RoBERTa showed the largest degradation, with accuracy falling below the 50% baseline for the balanced test set.

4.4 A Second Leakage Source: Subject Category

To determine whether additional shortcuts remained, all three models were retrained from scratch using dateline-stripped data. Surprisingly, performance remained extremely high, with DistilBERT achieving 99.72% accuracy and RoBERTa achieving 96.45%. This suggested the presence of another source of dataset-specific information.

Further analysis of the subject field revealed complete separation between subject categories and class labels. Real articles appeared only under politicsNews and worldnews, while fake articles appeared exclusively under the remaining six subject categories. A classifier using only the subject field, without access to article text, achieved 100% accuracy.

Although the subject field was not provided directly to the transformer models, the writing style, vocabulary, and structural characteristics associated with the different source categories may be implicitly encoded in the article text. Therefore, the models could potentially infer the class from textual characteristics correlated with the subject category rather than from the factual content of the articles.

4.5 Combined Controlled Test

To evaluate performance after removing both identified shortcuts, dateline stripping was applied to both the training and test data, and a subject-holdout split was used. The models were trained on four subject categories and evaluated exclusively on four previously unseen subject categories.

Model	Accuracy	vs. Majority Baseline (85.8%)
TF-IDF + Logistic Regression	92.29%	+6.5 pts
RoBERTa	86.80%	+1.0 pts
DistilBERT	21.91%	-63.9 pts

Table 4. Combined dateline-stripped + subject-holdout controlled test. Test set was imbalanced (85.8% real / 14.2% fake); majority-class baseline shown for reference.

The TF-IDF model retained meaningful predictive performance, including 93% recall on fake articles, while RoBERTa performed only slightly above the majority-class baseline. DistilBERT achieved 21.91% accuracy, substantially below the majority baseline. However, this result should be interpreted in the context of the distribution shift between the training and test sets. The training data contained approximately 58% fake and 42% real articles, whereas the held-out test set contained approximately 86% real and 14% fake articles. Consequently, a model biased toward the training-set majority class could perform poorly on the test distribution regardless of the amount of genuine content-level information it had learned. Therefore, the 21.91% result should be interpreted as evidence of severe shortcut dependence combined with distribution shift, rather than as definitive evidence that the model learned no genuine signal.

4.6 Discussion

The larger degradation observed for RoBERTa compared with DistilBERT in the dateline-only stripping experiment (Section 4.3) is consistent with findings from the shortcut learning literature. Higher-capacity models can more effectively exploit spurious but easily separable patterns when such signals are available, which can result in greater performance degradation when those patterns are removed (Geirhos et al., 2020).

However, this relationship was not consistent across the experiments. In the combined controlled test (Section 4.5), DistilBERT experienced a substantially larger performance decline than RoBERTa. This difference may be influenced by the class imbalance and distribution shift between the training and held-out subject categories discussed in Section 4.5, rather than representing a consistent difference in shortcut reliance between the two architectures. Therefore, the results do not support a definitive conclusion about whether model capacity alone determines sensitivity to the identified shortcuts. Both findings are reported to provide a complete view of the experimental results, and resolving this difference would require additional controlled experiments.

Overall, this investigation demonstrates that explainability methods can serve as practical model-auditing tools in addition to their conventional role in communicating predictions. In this project, SHAP and LIME helped identify influential features that led to the discovery of dataset artifacts and motivated controlled experiments to quantify their impact. This highlights the importance of examining model explanations when unusually high performance may be influenced by dataset construction, source characteristics, or other unintended correlations with the target labels.

5. Part 2 – LLM-Generated Explanations on LIAR

5.1 Pipeline

Given the leakage findings discussed in Section 4, the LIAR-trained DistilBERT model was selected as the primary model for the explanation pipeline. LIAR was preferred because it does not exhibit the same dataset-specific artifacts identified in ISOT, making it more suitable for evaluating explanation quality.

The pipeline consists of three main steps. First, DistilBERT produces a prediction and confidence score for each statement. Second, SHAP, using a masking-based text explainer, identifies the top five tokens contributing to the prediction, with each attribution indicating whether the token pushes the prediction toward the fake or real class. Third, these attributions are provided to Google Gemini through its free-tier API with an explicit instruction to generate a two-sentence, non-technical explanation grounded only in the provided tokens and their directions, without introducing external reasoning.

LIME (Ribeiro et al., 2016) was also applied to the same disagreement cases to provide a comparison between the two explainability methods. Attribution word overlap and generation time were compared to assess differences between SHAP and LIME.

5.2 Faithfulness Evaluation

To evaluate whether the generated explanations remained faithful to the underlying model, three complementary metrics were computed over 20 cases where the baseline and DistilBERT predictions disagreed.

- **Token coverage:** the proportion of SHAP's top-five tokens that appear verbatim in the generated explanation.
- **Directional accuracy:** whether the explanation correctly represents the SHAP-attributed direction of each mentioned token, either toward the **fake** or **real** class. Direction was assessed based on the surrounding wording and context of the explanation.
- **Consistency:** the stability of explanations when the same case is processed five times. Both lexical consistency, measured using word overlap, and semantic consistency, measured using sentence-embedding cosine similarity with *all-MiniLM-L6-v2*, were reported. Using both measures helps distinguish genuine changes in meaning from harmless differences in wording.

Metric	Result
Token coverage	78%
Directional accuracy	100% (15/20 checkable cases)
Lexical consistency (word overlap)	38%
Semantic consistency (cosine similarity)	0.88

Table 5. Faithfulness evaluation results on LIAR.

The difference between lexical consistency (38%) and semantic consistency (0.88) suggests that the LLM produced substantially different surface wording across repeated generations while generally preserving similar meaning. This distinction is important because low word overlap does not necessarily indicate that explanations are unfaithful when the same information can be expressed using different words.

Manual inspection of five repeated generations for a representative case, “Wisconsin is on pace to double the number of layoffs this year,” showed that all five generations identified the same influential tokens and reached the same overall conclusion, while varying primarily in their wording.

Overall, the results indicate that the generated explanations were semantically stable but lexically diverse. This suggests that the pipeline can produce varied, user-readable explanations while maintaining consistency with the model-attributed features. However, the evaluation is based on a relatively small set of cases, so these results should be interpreted as evidence of promising explanation stability rather than a definitive measure of general faithfulness.

6. User Study

6.1 Design

A user study was conducted with 12 non-technical participants to evaluate the readability and perceived usefulness of the generated explanations. Each participant reviewed 10 statements from the LIAR test set: five correctly classified statements, three incorrectly classified statements, and two cases where the baseline and DistilBERT predictions disagreed. Each statement was presented through the project's Streamlit dashboard along with the model prediction, confidence score, and LLM-generated explanation.

For each statement, participants provided two ratings on a 5-point Likert scale: clarity ("Did this explanation make sense?") and trust ("Does this explanation make you trust the prediction more?"). Participants could also provide an optional free-text comment. At the end of the study, participants were asked to choose between the plain-English explanation and a raw SHAP word-importance list.

6.2 Results

Example	Clarity M (SD)	Trust M (SD)
1	4.08 (0.79)	3.92 (0.90)
2	4.17 (0.58)	3.83 (0.94)
3	3.67 (0.89)	3.42 (0.79)
4	4.25 (0.62)	4.00 (0.60)
5	3.92 (0.79)	3.67 (0.78)
6	3.58 (0.79)	3.25 (0.97)
7	3.50 (0.67)	3.17 (0.72)
8	3.42 (0.90)	3.08 (1.00)
9	4.08 (0.51)	3.83 (1.11)
10	4.33 (0.49)	4.08 (0.67)
Overall	3.90 (0.76)	3.62 (0.90)

Table 6. Per-example and overall Likert ratings ($n=12$).

Preference	Participants	Percentage
Plain-English explanation	10	83.3%
Raw SHAP word list	1	8.3%
No strong preference	1	8.3%

Table 7. Explanation format preference ($n=12$).

6.3 Discussion

The explanations received an overall clarity rating of 3.90/5 and a trust rating of 3.62/5, both above the midpoint of the 5-point scale. In addition, 83.3% of participants preferred the plain-English explanations over the raw SHAP word-importance list. These results provide preliminary evidence that translating

feature attributions into natural-language explanations may make model outputs more accessible to non-technical users.

Examples 7 and 8 received the lowest clarity ratings (3.50 and 3.42, respectively) and corresponded to the two lowest model confidence levels (60% and 50%). This pattern suggests that lower-confidence predictions may be more difficult for users to interpret, although the small sample size does not allow us to conclude that prediction confidence and explanation clarity are statistically correlated. One implication for future interface design is to explicitly communicate model uncertainty alongside the explanation. For example, explanations associated with low-confidence predictions could include a short statement indicating that the model is uncertain, rather than presenting the generated explanation without additional context.

7. Dashboard

A Streamlit web application was developed to allow users to enter a headline or statement and receive a prediction, confidence score, top SHAP-attributed words, and a Gemini-generated plain-language explanation in real time. The SHAP-attributed words are color-coded based on whether they contribute toward the fake or real prediction. The dashboard uses the LIAR-trained DistilBERT model and the same SHAP-to-LLM pipeline evaluated in Section 5. It was also used to generate the materials presented to participants during the user study in Section 6.

8. Limitations

Several limitations should be considered when interpreting the results of this project.

- **User study:** The user study used a convenience sample of 12 primarily non-technical participants rather than a larger and more diverse participant group. The evaluation focused on clarity, trust, and explanation preference rather than task-based measures of understanding. A larger study involving different user groups could provide a more comprehensive evaluation.
- **Directional accuracy:** Directional accuracy was determined using a keyword-proximity heuristic rather than full semantic analysis and was checkable in only 15 of the 20 evaluated cases. The remaining cases contained ambiguous or absent directional language in the generated explanations.
- **Controlled ISOT evaluation:** The combined controlled test in Section 4.5 involves both shortcut removal and a class-imbalance distribution shift between the training and held-out subject splits. Therefore, the 21.91% DistilBERT accuracy should not be interpreted as definitive evidence that the model learned no genuine content-level signal.
- **SHAP tokenization:** For short LIAR statements, SHAP occasionally identified subword tokenizer fragments, such as parts of names split by the DistilBERT tokenizer, as important features. Stopword and punctuation filtering was used to reduce this issue in the user study, but it was not eliminated in all cases.
- **Cross-dataset generalization:** The models were trained and evaluated separately on ISOT and LIAR. A direct cross-dataset evaluation, such as training on ISOT and testing on LIAR without retraining, was not completed within the project timeline. Therefore, the project's evaluation does not measure how well the models generalize across different fake news datasets.

9. Conclusion and Future Work

This project developed a fake news classification system with SHAP-grounded, LLM-generated explanations and investigated the reliability of the underlying model predictions. The analysis of the ISOT dataset showed that its near-perfect benchmark performance was strongly influenced by two dataset artifacts: the Reuters dateline and the subject-category distribution. These artifacts were identified through explainability analysis of model predictions, disagreements, and false positives, demonstrating that SHAP and LIME can be useful not only for explaining predictions but also for identifying potential dataset issues.

The second part of the project developed an explanation pipeline using the LIAR-trained DistilBERT model, SHAP, and Google Gemini. The generated explanations achieved 78% token coverage, 100% directional accuracy on the checkable cases, and a semantic consistency score of 0.88 across repeated generations. In the user study, participants gave average ratings of 3.90/5 for clarity and 3.62/5 for trust, while 83.3% preferred the plain-English explanations over the raw SHAP word-importance list. These results provide preliminary evidence that LLM-generated explanations can make model predictions more accessible while remaining grounded in model-derived feature attributions.

Future work could include conducting a larger and more diverse user study with task-based measures of understanding, developing a more systematic evaluation of LLM explanation failure cases, and comparing different LLMs in terms of faithfulness, explanation quality, and computational cost. Cross-dataset transfer experiments could also be conducted to further evaluate the generalization of fake news detection models across different datasets. These extensions would provide a more comprehensive evaluation of both the classification models and the proposed explanation pipeline.

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